

Design for Serendipitous Interaction: BubbleBot - Bringing People Together with Bubbles

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Abstract—Fast-paced contemporary life full of planned interaction usually makes people miss out on wonderful moments. We here present BubbleBot, a speculative robot designed to support serendipity of interactions in public space. After observations in public spaces and embodied design workshops, we have designed BubbleBot to be a peripheral public-space robot, bursting bubbles at passersby to invite serendipitous interactions. BubbleBot is a speculative robot to create magical moments among people with minimal peripheral social interaction. With this project, we aim at generating a conversation about the future roles and interaction paradigms of robots in public space.

Index Terms—Serendipitous Interaction; Interaction Design; Public-space Robot

I. INTRODUCTION

Contemporary urban life is fast paced: we run from one appointment to another, strive to navigate the environment effectively and efficiently. The current design space of public space and personal communication technology appear not to sufficiently afford human-human social encounters, especially casual social interactions that bring about serendipity [1], [2]. Serendipitous interaction, casual chats with not acquainted, have long been judged as having positive social impacts favoring group interaction [3]. In this project, we venture the design space for serendipitous interaction in public space designing a peripheral robot, BubbleBot, shown in Fig. 1.

The design space of robotic technology in the public space mainly revolves around supporting efficiency, effectiveness, and personal space. Personal mobile devices are designed to made it easier to form private individual spaces in public domain. As this phenomenon is gradually transforming public spaces to be less interactive, researchers are working on using technology oppositely as the facilitator for bringing public interaction back [4]. We here propose the idea of using robots as the means to slow down the fast-paced urban life and creating serendipitous interaction.

Robots have been proven capable of influencing humans interpersonal behaviours and might be a possible application for creating serendipitous interactions among people [5]. Physical embodiment and social affordances have placed robots in a unique position of potential anchor for social interaction in public space. Many researchers and designers have explored the design of robots for social interactions in public space [6], but none (at the best of our knowledge) have explored



Fig. 1. The original design of BubbleBot.

serendipitous interaction. We here explore the design space of social and serendipitous interaction in public space with a speculative design approach [7], combining imaginative, playful thinking with existing interaction design approach, the theory of implicit interaction [8] and the embodied design approach [9].

Observing how making soap bubbles in public space bring people together, we tried to replicate the same experience with a peripheral mobile robot (i.e., not in the focus of attention) that burst bubbles at passersby to nudge towards serendipity. In this paper, we present our concept, design process and robot design. We aim to finalize the design informed by the in-the-wild observations and to have two prototypes to deploy in North America and in Europe that will inform next design iterations. Our project aims not just to create a product but to generate ideas, discussion and lay the groundwork for design project of robot for social and serendipitous interactions.

II. DESIGN CONTEXT

For this project, we are targeting to induce short chats among passersby in public space and for them to slow down the fast pace and enjoy the moment. We hope with our design, people can get to experience serendipitous engagement with one another. Following this objective, three design guidelines were defined as the goal for our design process.

- (1) **Peripheral** - The robot is meant to work in the periphery of the human-human interaction as we aim at increasing serendipitous interactions among passersby.
- (2) **Joint-attention** - The robot should be able to catch joint-attention and slow down passersby's fast pace and further create the desired serendipitous interaction later.
- (3) **influential** - The robot should induce the willingness of passersby to interaction with others, slightly nudging passersby's behaviour towards serendipitous interactions.

III. DESIGN PROCESS

We conducted observations in several public spaces to seek for serendipitous interaction. One thing that caught our interests was the activity of blowing bubbles - bubbles can bring people together as they not only draw the joint-attention, but also provoke pointing and poking behaviours. We came up with the initial idea of a mobile bubble robot that utilizes the features of bubbles to influence humans behaviours.

To **dip** deeper into the idea, we held a small workshops inviting people to come share the images they have for bubbles. People seemed to feel relaxed and happy when being around bubbles as someone mentioned: bubbles give people the opportunity the act like children. Another interesting comment is: whenever I saw bubbles, I cant help but start poking them. We hope to replicate all these wonderful images people have for bubbles with BubbleBot.

Another workshops was held for **embodied design** improvisation in which designers and researchers prototyped the appearance and interactions for BubbleBot. We started our first design of the robot from a minimal and peripheral aspect and planned out the pattern for interactive behaviours. We are planning to conduct observations of the robot in-the-wild to see peoples reactions toward BubbleBot in order to inform next design iterations.

IV. SCENARIO AND BEHAVIOUR

The preset scenario for BubbleBot is to **wander around public space, burst bubbles toward passersby, and create the opportunity for them to start an interaction with each other. We want to infuse the typical images and feelings that people have for bubbles into the behaviours of BubbleBot as creating a pleasing environment.** Four **sequential** behaviours for BubbleBot were designed to initiate interaction, shown in Fig. 2:

- (a) **Approaching** - moving forward to people whom BubbleBot is interested in interact with.
- (b) **Blowing Bubble** - drawing joint attention and showing interests.
- (c) **Playing with Bubbles** - blowing more bubbles to elicit poking and pointing behaviours toward the bubbles.
- (d) **Leading with Bubbles** - leading people to others with bubbles in the hope to initiate interactions.

V. ROBOT DESIGN

The original design of our BubbleBot is shown in Fig. 1. We wanted to embody and elicit the senses of **playfulness and friendliness** in the appearance of it. We minimized the

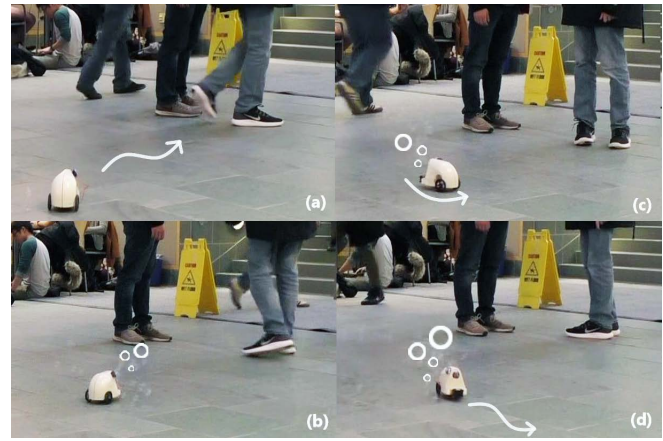


Fig. 2. Illustration of BubbleBots sequential behaviours.

size of the robot and chose to use vivid colors to make it less intimidating and more approachable. The interior design of BubbleBot includes the bubble-blowing mechanism and two DC motors all controlled by an Arduino Mini. Since this design is still a trial version, we chose to apply Wizard-of-Oz technique to control the robot with Bluetooth module through developed mobile phone app interface.

VI. FUTURE PLAN

In the future, with the designed behaviours of BubbleBot, we are planning to **try out the planned scenario in several public places** in North America and Europe, as we are aware of possible cultural difference. We are also planning to add more features, for example, **the ability to print out messages, to expand the social affordance of BubbleBot and last the serendipitous interactions.** We will **perform iterations** for the design of BubbleBot to demonstrate it during the conference.

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